

silient insect called the fruit and shoot borer (FSB) that eats into tender shoots and fruits retarding growth, making the brinjals unsuitable for the market and unfit for human consumption. Heavy doses of pesticide just didn't seem to help.

That's when Maharashtra Hybrids Seed Company (Mahyco) combined forces with the US seed giant Monsanto to come up with a GM brinjal variety that would become resistant to the borer. Both these companies had in the past jointly developed the genetically modified Bt Cotton to successfully tackle the bollworm problem that had devastated cotton crops in the past. They did this by introducing into the cotton seed a gene of the common soil microbe called *Bacillus thuringiensis* that encoded an insecticidal protein lethal to the bollworm hence the name Bt Cotton. When the companies found that the Bt gene was as effective in tackling the brinjal borer they decided to develop the transgenic Bt brinjal.

Estimates are that Bt brinjal could add to the current annual production of 80 lakh tonnes by more than 50 per cent—that is as much destroyed by pests—with Bt brinjal. This offers a win-win situation both for farmers and consumers. Brinjal is grown in around 5.5 lakh hectares and is a critical cash crop for more than 1.4 million small and marginal farmers. The area under cultivation has gone up by 15 per cent in the last 10 years but production has barely increased because of repeated borer attacks. Sadly, despite a range of brinjal varieties, India, which is also the second largest producer in the world after China, has not been able to check the virulence of the FSB that is responsible for the pockmarks on the vegetable's peel. "It will help millions of brinjal farmers who have been suffering from the havoc caused by FSB and it will help farmers tackle this pest in an environment friendly manner and increase yields and farm income," says Raju Barwale, managing director, Mahyco.

“The GEAC is hand-in-glove with the industry and is not on the side of the public. It has failed the country.” RAJESH KRISHNAN, *Greenpeace campaigner*

THE GM DECODER

It's high on science but public credibility continues to be low

Transgenic Crop

A transgenic crop is one which contains a gene or genes of a different species artificially inserted in its genome, which may come from an unrelated plant or from a completely different species.

What's so special about it?

A transgenic plant carries new gene(s)

New gene(s) can be introduced into a plant from any source—bacteria, virus, fungi, animals—to improve yields and make it resistant to its pests.

Biotechnology is the new age version of plant breeding

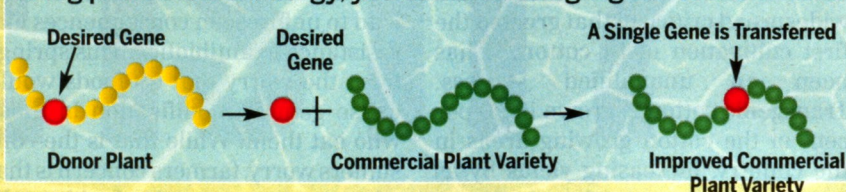
TRADITIONAL PLANT BREEDING

DNA is a sequence of genes that determines the traits of a plant. Traditional plant breeding uses many genes to improve quality.



PLANT BIOTECHNOLOGY

Using plant biotechnology, you can add a single gene to the strand.



About safety concerns, Mahyco points out that the company fulfilled the series of trials both for its efficacy and safety including the toxicity tests on animals that the Genetically Engineering Approval Committee, (GEAC) of the Union Ministry of Environment and Forests had ordered. Two expert committees formed by the GEAC had extensively reviewed the bio-safety data and had cleared Bt brinjal after large-scale field trials were done in public sector

research laboratories. While the Indian Council of Agricultural Research did multi-location agronomic trials in 2004 and 2005, the Indian Institute of Vegetable Research did large-scale trials in 2007 and 2008. "Rigorous scientific tests, including toxicity and allergenicity evaluation as well as nutritional studies have confirmed that Bt brinjal is as safe as its non-Bt counterpart," claims Mahendra Kumar Sharma, Mahyco's general manager.

Such assertions do not wash well with civil society groups and NGOs who have vehemently oppose the intro-